

6.8 Case Study: Digital Inquiry Project

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟"

CS XI

Chapter 6: Digital Literacy

Lecture 10 of 14

★ Golden

6.8 Case Study: Digital Inquiry Project ★ Golden

LEARN IT

This case study shows how to apply **all** the ideas of this chapter to solve a real-world problem from start to finish. An **inquiry** is an investigation to find the answer to a question.

Research question: "How does daily mobile phone use affect the study habits of students?"

Scenario: Students notice classmates using phones during free periods and while studying. Some say phones **help** them learn; others say phones **waste time**. The class must investigate whether phone use helps or harms study habits.

The digital inquiry workflow

1 Question



2 Search



3 Collect



4 Analyse



5 Create

The project is carried out in **12 steps** that fit into this five-stage workflow:

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data
Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

REMEMBER IT

Workflow in five words: **Ask → Search → Collect → Analyse → Create**. Every research project you ever do will follow this pattern.

اردو میں سمجھیں

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟" تحقیق پانچ مراحل میں ہوتی ہے: سوال، تلاش، ڈیٹا جمع کرنا، تجزیہ اور آخری پروڈکٹ بنانا۔ یہ کام بارہ قدموں میں مکمل کیا جاتا ہے۔

EXAM TIP

Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps with one or two lines each, and finish with the artefact. Learn the five-stage workflow as your skeleton.

CHECK YOURSELF

1. What is the research question in the case study?
2. Name the five stages of the digital inquiry workflow.

Answers: 1. "How does daily mobile phone use affect the study habits of students?" 2. Question, Search, Collect, Analyse, Create.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The digital inquiry project applies:

- a. All of chapter 6 in order
- b. Only careers
- c. Only NAND
- d. Only OSI

MCQ 2. Advanced search helps you:

- a. Narrow secondary sources (quotes, site, date)
- b. Hack Wi-Fi
- c. Draw K-maps
- d. Sort in $O(1)$

MCQ 3. Primary collection in the project is often:

- a. A class survey
- b. A CPU design
- c. A TCP stack
- d. A waterfall contract

MCQ 4. You organise results in a:

- a. Spreadsheet + charts
- b. Random WhatsApp dump only
- c. NAND netlist
- d. Figma only

MCQ 5. The artefact is:

- a. The final digital product that answers the question
- b. The first brainstorm sticky
- c. The career list
- d. The teacher's attendance

MCQ 6. A conclusion should:

- a. Answer the question and admit limits
- b. Ignore the data
- c. Name and shame students
- d. Invent $n=1$ million

Extra short questions

1. List the project steps in order (grouped as in the book).
2. What is a digital artefact here?
3. Why include secondary if you surveyed?
4. One sentence of a good limitation.

Extra long question

1. Outline a 12-step inquiry: "Does extra screen time lower concentration in our class?" Name tools and the artefact. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Search/method; survey/primary; secondary+sheet; analyse/conclude; artefact.
2. Slides, infographic, short video, site — that answers the inquiry question.
3. To compare your small sample with a published claim.
4. $n=40$ in one college, so we cannot speak for all Sindh.

Long

1. Steps 1–3 search; 4–6 survey; 7–9 PBS/article + sheet; 10–12 charts, conclusion, artefact (infographic+spreadsheet). Ethics. One expected chart: hours vs self-reported focus.

Next lecture: Step 1 – 3: Advanced Search & Methodology